RESTORATION AND ENHANCEMENT OF COVID-19 VARIENTS USING CT IMAGES

Ranjani.R [1] and R.Priya [2]
[1] Ph.D Scholar, Vels Institute of Science, Technology and Advanced Sciences (VISTAS)
[2] Professor, Vels Institute of Science, Technology and Advanced Sciences (VISTAS)

[1]ranji010794@gmail.com
[2]priya.scs@velsuniv.ac.in

Abstract Covid (COVID-19) is an irresistible illness brought about by the SARS-CoV-2 virus. Initially it was first found in China, and began to spread quickly all around the world causing numerous deaths and defeats in practically all fields. Computed tomography (CT) images are popularly being used in the field of computer vision to aid the medical experts to diagnose various diseases. This work aims to pre-process raw CT images which contains noises and disturbances which needs to be filtered and enhanced for various medical applications .In this paper, we propose pre-processing steps (restoration and enhancing) of its variants utilizing CT scan images. Lung computerized tomography (CT) images can be viably utilized for early identification of Covid-19 patients. These CT images are handled utilizing Computer Aided Diagnosis (CAD) procedures by the use of reasonable calculations. The first and fundamental stage for any sort of image is to restore and improve them. In this paper the restoration algorithms used is a combination of two traditional algorithms (bilateral+anisotropic) thus naming it Improved Anisotropic Diffusion Bilateral Filter (2D IADBF). The efficiency of the proposed techniques is exhibited through comparison between traditional algorithms like 2D Median Filter, 2D Log Filter and 2D Frequency Domain Wavelet Filter for restoration. The enhancement algorithm utilized is 2D Edge Preservation Efficient Histogram Improvement (2D EPEHI) algorithm which is ensemble of Edge preservation and histogram processing which focuses on methods like Contrast Limited Adaptive Histogram Equalization (CLAHE), 2D Adaptive Mean Adjustment and Image Coherence improvement and the results are efficient.

Keywords: Covid-19, variants, enhancement, restoration, CAD, CT image.

1 Introduction

Computed tomography (CT) images are popularly being used in the field of computer vision to aid the medical experts to diagnose various diseases. The CT images of the lungs can be used for the diagnosis of multiple diseases including lung cancer and Covid-19[1]. The Covid-19 viruses have taken multiple forms ever since it was first discovered. Viruses constantly change due to mutation. When a virus undergoes one or more mutations, it is called a variant. Coronavirus has caused several variants across the globe. Once a virus enters the physical body, its genetic material enters the cells and starts making copies of it which may infect the other cells. Whenever a blunder occurs during this copying process, it triggers a mutation.

Early discovery of Covid-19 aids in the ideal arrangement of clinical consideration to the patients. This recognition is possible with the assistance of lung CT images. There are two fundamental ways for the discovery of Covid-19[6].The principal classification is the lab based methodology. This approach includes the examination of tests of human like bodily fluid, throat swabs, and blood. These samples are exposed to testing which includes nucleic acid testing, antigen testing, point testing, serology testing, etc. [7]. In these tests, the swab materials are tested over paper strips that have counterfeit antibodies. The normal awareness of this technique is around 60-71%. The subsequent classification is the use of clinical imaging methods like CT examine, x-ray, and so forth The raw CT images acquired are applied to imaging methods and are exposed to different algorithms[8].The algorithms used in this work gives imaging information and give efficient results .

The primary methods of Image processing are pre-processing, segmenting and clustering. The pre-processing includes steps like noise removal and enhancement [9]. Restoration is the initial step utilized for the evacuation of noise that is consolidated into the image during the image acquisition step. The image is enhanced and utilized for working on the Brightness and differentiation elements of the image. Proper plan of filtering and improvement algorithms are required to work on the exactness of infection diagnosis [10].

Subsequently, in this research, two novel calculations are proposed for the image filtration and enhancement of the CT images to analyses Covid-19. Covid-19 variants are divided into three types. They are: Variant of Interest, Variant of Concern and Variant of High Consequence. The table1 illustrates the different kinds of variants with its origin, symptoms and countries of spread.

Table 1. Origin and Symptoms of Covid-19 Variants

Variant	Originated from	Month and year	Countries affected	Symptoms
Alpha Variant	United Kingdom	September 2020	Minimum of 173 countries such as the United Kingdom, Bulgaria, Wales, Ireland, United States.	Chills, loss of appetite, headache and muscle aches.
Beta Variant	South Africa	May 2020	122 countries including United States	50% more transmissible, severe infections.

Gamma Variant	Brazil	November 2020	62 countries including Canada, Italy, United States	Body aches, sore throat, diarrhoea, conjunctivitis, and headache, loss of taste or smell.
Delta Variant	India	2020	96 countries including Singapore, France, Spain, UK, USA	Persistent cough, headache, fever, and sore throat. Headache, sore throat, runny nose, and fever seem to be more common.
Omicron variant	South Africa	2021	Minimum 50 countries which include south Africa , USA, Canada, Ireland, Australia etc	start with body ache, generalised weakness, fatigue, headache and fever, cough /cold where there is water from the nose, sneezing, etc..

2 Literature survey:

Ref.Paper & Year	Authors	Purpose	Author's Proposed system
1. "Imaging manifestations and diagnostic value of chest CT of coronavirus disease 2019 (COVID-19) in the Xiaogan area," *Clinical . Radiology., Volume 75, Issue 5, May 2020*, vol. 75, no. 5, pp. 341–347, 2020, *Elsevier.*,doi: 10.1016/j.crad.2020.03.004.[11]	K. Wang, S. Kang, R. Tian, X. Zhang, and Y. Wang,	To Diagnose of covid-19 using chest images	To test the relationship between severity level of infection and lymphocyte ratio. To test the oxygen level of patients
2., "Diagnosis of the Coronavirus disease (COVID-19): rRT-PCR or CT?," *European Journal of Radiology.*, vol. 126, no. March, p. 108961, *2020, Mar 25, National library of medicine* .,doi: 10.1016/j.ejrad.2020.108961.[12]	C. Long *et al.*	To compare the diagnostic values of CT scans and RCT-PTR Test	Detection of covid-19 based on CT Real-time reverse transcriptase-polymerase chain reaction(RCT-PTR Test)
3., "COVID-19 pneumonia: A review of typical CT findings and differential diagnosis," *Diagnostic and Interventional Imaging, Volume 101, Issue 5, May 2020, Pages* 263-268, *Elsevier.*,doi: 10.1016/j.diii.2020.03.014.[13]	C. Hani *et al.*	To find a review on CT findings for covid-19 case	Analysis of various microbiological tests in identification of covid-19 like RCT-PTR Test and sequencing tests
4. "Diagnostic performance between CT and initial real-time RT-PCR for clinically suspected 2019 coronavirus disease (COVID-19) patients outside Wuhan, China," *Respiratory Medicinej.jul 2020* vol. 168, no. April, p. 105980, 2020, *National library of medicine* .,doi: 10.1016/j.rmed.2020.105980.[14]	J. L. He *et al*	To compare the diagnostic values of CT scans and RCT-PTR Test	The outcomes had no statistical difference between the 2 methods
5. "Lung X-Ray Segmentation using Deep-CNN on Contrast-Enhanced Binarized Images", MDPI, Mathematics, June 2020[18]	Hsin-Jui Chen And Yan-Tsung Peng	Automatic Locating of Lung Regions in CXR Images is important on CAD	Applying other image enhancement and binirization methods for training coverage

The literature survey obviously shows that chest CT is exceptionally compelling in the acknowledgment of Covid-19. Further, it was seen that the awareness achieved utilizing chest CT is higher than that of continuous converse transcriptase-polymerase chain response. Along these lines, chest CT information is utilized for the locationCovid-19 in this research.

3 Existing Method and its Limitations :

The proposed methods for image filtration utilize either anisotropic diffusion or bilateral filtering separately. In this work the proposed algorithm works as a combination of both frequency and spatial domain filtering and thus simultaneous conservation of gradient data alongside the speckle noise (a type of noise found in images) decrease is impossible. Further, the existing enhancement methods that perform histogram leveling bring about the blurring of the image because of the deficiency of edge data.

4 Proposed method and Advantages:

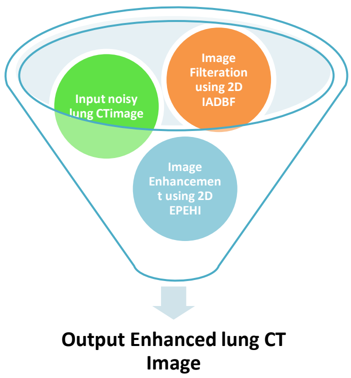


Fig. 1. Illustration of Proposed Method

4.1 2D Improved Anisotropic Diffusion Bilateral Filter (2D IADBF)

Image restoration is done to filter out the speckle noise from the lung CT images. This is done based on the combination of anisotropic diffusion model and the bilateral filter and was implemented using 2D Improved Anisotropic Diffusion Bilateral Filter (2D IADBF).The Restoration algorithm is a combination of Spatial and Frequency domain making it an Spacio-Frequency Domain (i.e. Bilateral + Anisotropic).The Anisotropic algorithm is a spatial domain algorithm which removes noise and preserves gradient information.

Similarly the Bilateral Algorithm (Frequency domain) is used in speckle noise detection is very high. To achieve the improved variation form of filtering is the 2D Improved Anisotropic Diffusion Bilateral Filtering algorithm is used.. The algorithm is depicted below:

Algorithm 1: 2D Improved Anisotropic Diffusion Bilateral Filter (2D IADBF).

Input:.

Input noisy CT lung image $I^{CT} \in R^{p \times q}$.

Steps: .

For the given input image define the anisotropic diffusion model using Equation (1).

The diffusion coefficient is generated based on the gradient function using $DC^i(u,v) = gd(\Delta I)$.

The gradient function is modified using bilateral function as $gd(\Delta I) = \dfrac{1}{1+(|\Delta I|/BF)^2}$.

The bilateral function is defined using the gradient function to enable accurate noise filtration using Equation (4).

Finally, the filtered image $F^{CT} \in R^{p \times q}$ is obtained using,

$$F^{CT}{}_{t+1}(u,v) = F^{CT}{}_{t}(u,v) + \frac{1}{4}[DC^i(u,v).\sigma^2(u,v)]$$

Output:

Filtered image $F^{CT} \in R^{p \times q}$.

4.2 2D Edge Preservation Efficient Histogram Improvement (2D EPEHI)

The next stage in the proposed system is the image improvement/enhancement which is oriented towards the general contrast and brightness of the image. The Enhancement algorithm is utilized for a powerful edge protecting and an effective histogram giving a better outcome. The algorithm for enhancement is depicted below.

2: 2D Edge Preservation Efficient Histogram Improvement (2D EPEHI).

Input:.

Filtered image $F^{CT} \in R^{p \times q}$.

Steps:.

- Divide the filtered image $F^{CT} \in R^{p \times q}$ into sub-blocks of size $n \times n$.
- For each sub-block, compute the Tenengrad criterion using Equation (6).

- Calculate the overall Tenengrad criterion for a particular block using $TCB = \sum_{u \in w} \sum_{v \in w} [TC(u,v)]^2$.
- Divide the image into 4 regions based on the Tenengrad criterion.
- Compute the weight function for each histogram using $wf(hm_i) = C(i) / MC$.
- Compute the modified histogram using $MH = \sum_{i=1,2,3} wf(hm_i) * hm_i$.
- Find the enhanced image $E^{CT} \in R^{p \times q}$ using the modified histogram.

Output:.

Enhanced image $E^{CT} \in R^{p \times q}$

5 Experimental setup

All the experiments performed in this research were accomplished using MATLAB R2013b with Intel core i5 processor @4 GHz having8GB RAM.

6 Results and Discussion

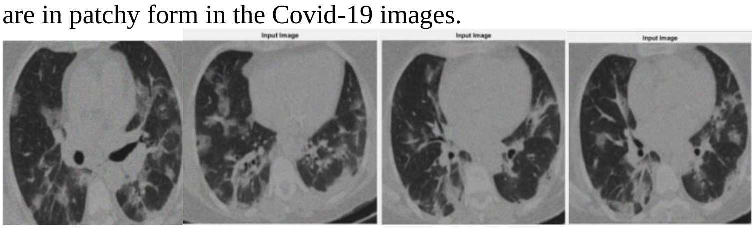
The proposed framework is implementation and results are discussed in this section.

7 Parameter Settings

The Covid-19 variant's lung CT images were obtained from the publicly available Imaging Archive dataset (https://www.mdpi.com/2076-3417/11/24/11902/html). The size of the sub-block $n \times n$ was chosen as 8×8.

8 Qualitative analysis

The quantitative analysis of the proposed system was performed using 4 types of variant images i.e. alpha, beta, gamma and delta. These variant images are shown in Figure 2. From the figure it is evident that both the images have lesions around the periphery regions. However, the shape of the lesions varies in the images. The lesions are in patchy form in the Covid-19 images.



(a) (b) (c) (d)

Fig. 2. (a)Alpha variant (b) Beta variant (c) Gamma variant (d) Delta variant

8.1 Qualitative analysis on Covid-19 images

Figure 3 a shows the Alpha variant Covid CT lung input image filtered using2D median filter (MF), 2D Adaptive Log Gabor filter (LOG), 2D Frequency domain wavelet filter (FDWF) and the proposed 2D IADBF filter. It is evident from Figure 3 that, the image is blurred in the case of 2D median filter. Similarly, the noise components are still prevalent in the 2D Adaptive Log Gabor filter. The contrast of the image filtered using 2D Frequency domain wavelet filter is also poor. However, the proposed 2D IADBF filter produces best filtration results overcoming all the issues in the other filters.

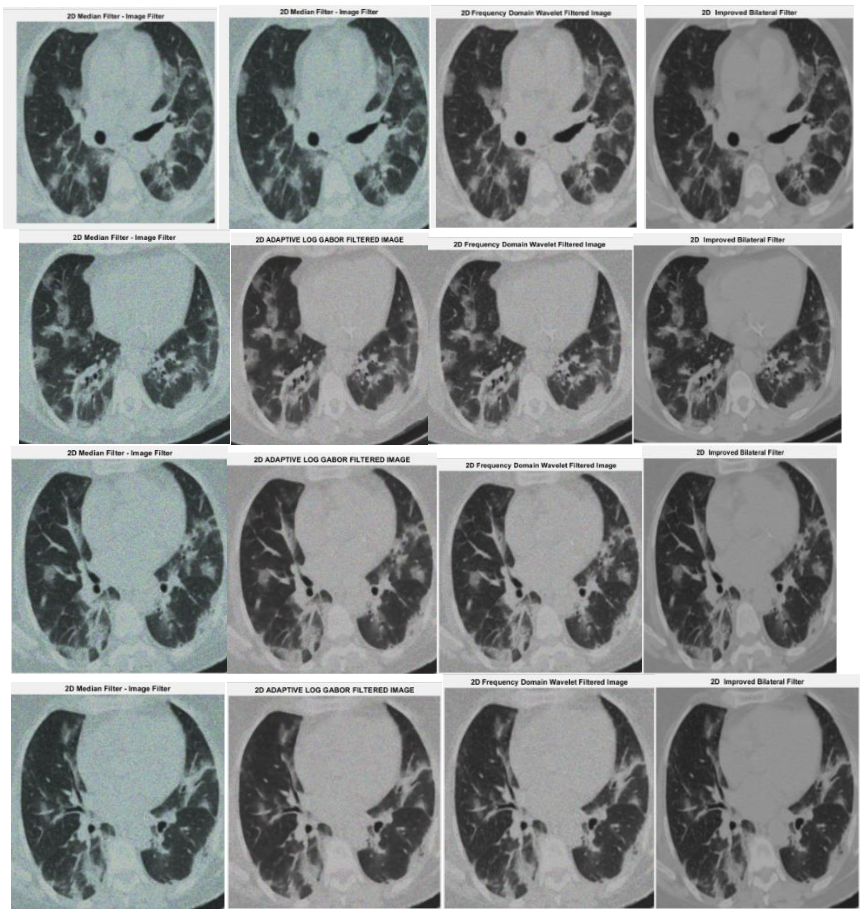


Fig. 3. (a) Alpha variant restorations (b) Beta Variant Restorations (c) Gamma variant Restorations and (d) Delta variant Restorations

Figure 4 shows the Covid-19 variant's CT lung filtered image enhanced using CLAHE, 2D AMA, Coherence adjustment and the proposed 2D EPEHI algorithm. The proposed 2D EPEHI algorithm achieves best enhancement results in which the brightness and contrast components are distributed uniformly throughout the image. However, in the other images, the presence of noise and blurring effect is prominent.

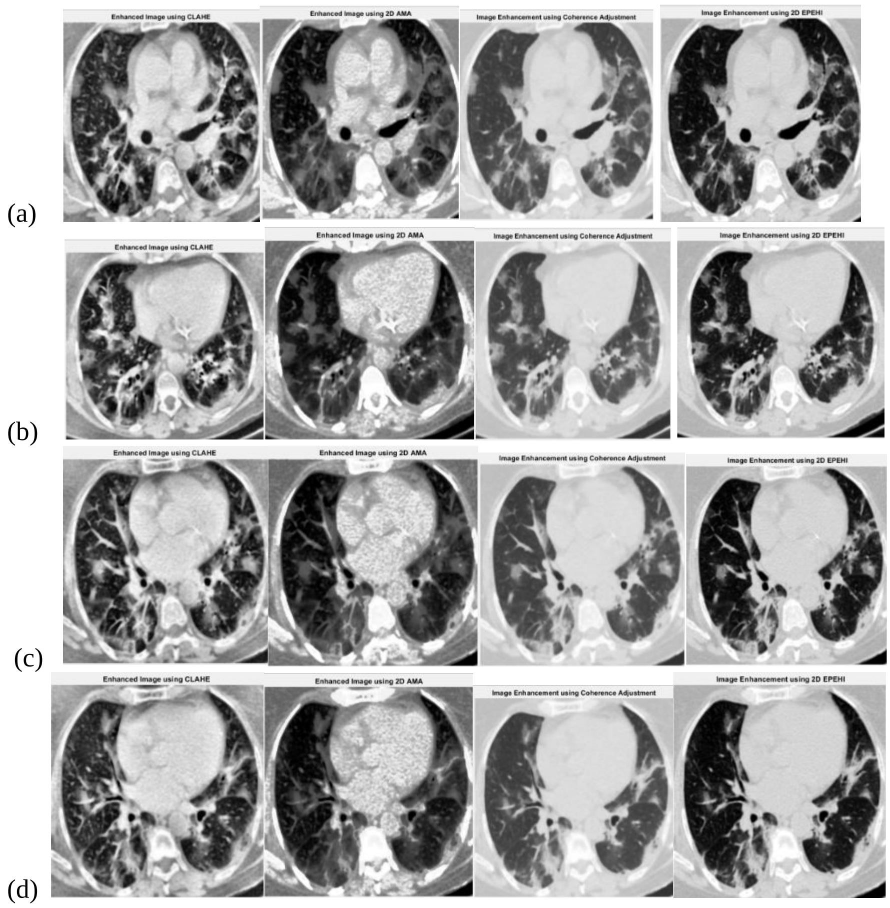


Fig. 4. (a) Alpha variant enhancements (b) Beta Variant enhancements (c) Gamma variant enhancements and (d) Delta variant enhancements.

Graphs of Covid-19 Variants.

Figure 5 shows the comparison of processing time graph(x-axis- Diff algorithms ; y-axis- processing time), RMSE graph(x-axis-Iterations :y-axis-Mean sqr values) and

PSNR graph(x-axis-Algorithms; y-axis-PSNR in dB) for different filtering algorithms using Covid-19 variant's CT lung image

(a

b

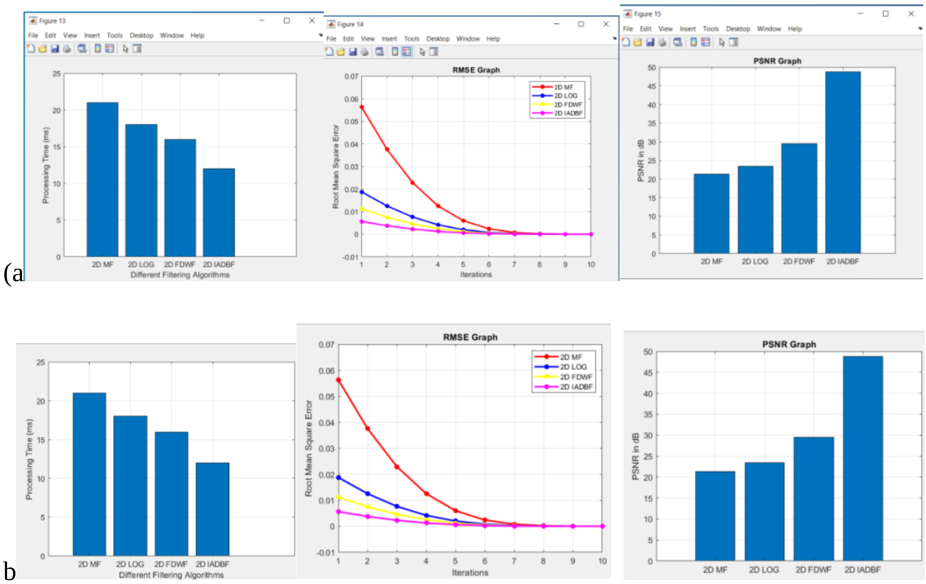


Fig. 5. (a) Alpha variant graphs (b) Beta Variant graphs

Table 2. Comparison tabulation of Covid-19 Variant's for Restoration Algorithms

Algorithm/ Variant	Parameter (PSNR in dB)	2D AMA	2D LGF	2D FDWF	2D IADBF
Alpha variant	PSNR	20.35	27.93	19.98	38.75
	MSE	59.94	10.46	65.29	0.85
Beta variant	PSNR	19.84	27.82	19.52	37.84
	MSE	67.82	10.72	72.50	0.83
Gamma	PSNR	19.95	27.88	19.60	38.00
Variant	MSE	65.70	65.70	71.22	0.81
Delta variant	PSNR	19.85	27.89	70.91	38.01
	MSE	65.65	10.56	19.60	0.81

The above tabulation clearly shows that the proposed algorithm's PSNR values of different covid variants is usually of the range 35-38 which means the image is signaled and removed noise in the specific range(i.e. the more PSNR more the image is noise free)and is measured in dB .Similarly the MSE values of the proposed algorithms of different variants show that there almost no error in the image i.e.

approximately of the range greater than 0.80-0.85.Note that MSE is a statistical value and does not have any units.

Table 3. Comparison Tabulation of covid-19 variant's Enhancement

Algorithm/ Variant	Parameter	CLAHE	2D AMA	CIA	2D EPEHI
Alpha variant	SSIM	0.25	0.38	0.31	0.44
	AMBE	10.42	6.96	6.96	4.72
Beta variant	SSIM	0.19	0.41	0.38	0.43
	AMBE	9.48	5.93	7.23	4.70
Gamma variant	SSIM	0.38	0.42	0.35	0.48
	AMBE	7.52	5.46	6.8	0.71
Delta variant	SSIM	0.49	0.44	0.32	0.42
	AMBE	8.59	5.32	6.63	4.74

The table 3 shows that the proposed algorithm's SSIM values of different covid variants is approximately of the range 0.40-0.44 which means the image is preserved with gradient and edge information in the specific range i.e. the more SSIM more the image is preserved .Similarly the AMBE values of the proposed algorithms of different variants show that image's average mean brightness error is approximately of the range 4.5-4.7.Note that both SSIM and AMBE are statistical values and does not have any units.

9 Conclusion and Future work

In this research, novel algorithms are proposed for the pre-processing of medical lung CT images. A new algorithm called 2D Improved Anisotropic Diffusion Bilateral Filter (2D IADBF) was proposed for the effective filtration of CT images. This scheme was designed such that the noise components of the image were completely removed with the simultaneous retention of gradient details for all the covid-19 variants. This system was modeled based on the union of bilateral filter and the anisotropic diffusion model. Further, a new scheme for image enhancement called 2D Edge Preservation Efficient Histogram Improvement (EPEHI) is also proposed. Thus the proposed algorithm achieved uniform distribution of components with minimal brightness error for all the 4 covid-19 variants. The proposed algorithms were evaluated using various evaluation metrics. It was inferred that the proposed 2D IADBF filtration algorithm achieved minimal processing time of 13ms for Covid-19 respectively. Further, the RMSE of the 2D IADBF filtration algorithm was as low as 0.80 for the Covid-19 CT images respectively. The proposed EPEHI algorithm attained high SSIM of 0.44 respectively. In future, we can design algorithms for the segmentation and classification of these images for the effective diagnosis of Covid-19 Variants which can be used in medical applications.

10 References

1. R. Schofield ., “Image reconstruction: Part 1 – understanding filtered back projection, noise and image acquisition,” Journal of Cardiovasular Comput Tomography, vol. 14, no. 3, pp. 219–225, 2020,*Elsevier*.
.
2. D. Racine ., “Task-based characterization of a deep learning image reconstruction and comparison with filtered back-projection and a partial model-based iterative reconstruction in abdominal CT: A phantom study,” Physica Medica, vol. 76, no. June, pp. 28–37, 2020, Elsevier.

3. Y. Deng *et al.*, “Usefulness of CT texture analysis in differentiating benign and malignant renal tumours,”Clinical Radiology. Volume 75, Issue 2, February 2020, Pages 108-115, Elsevier.

4. I. Ozsahin, B. Sekeroglu, M. S. Musa, M. T. Mustapha, and D. Uzun Ozsahin, “Review on Diagnosis of COVID-19 from Chest CT Images Using Artificial Intelligence,” Computational and Matematical Methods in Medicine., vol. 2020, Sept 26,2020,Hindawi.

5. A. Kumar, P. K. Gupta, and A. Srivastava, “A review of modern technologies for tackling COVID-19 pandemic,” Diabetes Metabolic Syndrome and Clinical Reservation.,National library of medicine, Revised Jul-Aug 2020;14(4):569-573.., vol. 14, no. 4, pp. 569–573, 2020, doi: 10.1016/j.dsx.2020.05.008.

6. A. Bhandary , “Deep-learning framework to detect lung abnormality – A study with chest X-Ray and lung CT scan images,” Pattern Recognition. Letters., vol. 129, pp. 271–278, jan 2020, Elsevier, .2019.11.013.

7. M. Toğaçar, B. Ergen, and Z. Cömert, “Detection of lung cancer on chest CT images using minimum redundancy maximum relevance feature selection method with convolutional neural networks,” Biocybernetics. Biomedical. Engineering., Volume 40, Issue 1, January–March 2020, Pages 23-39 2020, Elsevier.,

8. P. Mohamed Shakeel, M. I. Desa, and M. A. Burhanuddin, “Improved watershed histogram thresholding with probabilistic neural networks for lung cancer diagnosis for CBMIR systems,” Multimedia. Tools Applications., vol. 79, no. 23–24, pp. 17115–17133, 2020,Springer.,

9. L. Zhang *et al.*, “Clinical characteristics of COVID-19-infected cancer patients: a retrospective case study in three hospitals within Wuhan, China,” Annals of Oncology:official European jouranl of oncology., National library of medicine, vol. 31, no. 7, 2020 Jul;31(7):894-901.,pp. 894–901, 2020.

10. K. Li *et al.*, “CT image visual quantitative evaluation and clinical classification of coronavirus disease (COVID-19),” European. Radiology.,25 march 2020 vol. 30, no. 8, pp. 4407–4416, 2020,Springer,.doi: 10.1007/s00330-020-06817-6.

11. K. Wang, S. Kang, R. Tian, X. Zhang, and Y. Wang, “Imaging manifestations and diagnostic value of chest CT of coronavirus disease 2019 (COVID-19) in the Xiaogan area,” Clinical . Radiology., Volume 75, Issue 5, May 2020, vol. 75, no. 5, pp. 341–347, 2020, Elsevier.,

C. Long *et al.*, “Diagnosis of the Coronavirus disease (COVID-19): rRT-PCR or CT?,” European Journal of Radiology., vol. 126, no. March, p. 108961, 2020, National library of medicine .,

12. C. Hani *et al.*, “COVID-19 pneumonia: A review of typical CT findings and differential diagnosis,” Diagnostic and Interventional Imaging, Volume 101, Issue 5, May 2020, Pages 263-268, Elsevier.

13. J. L. He *et al.*, “Diagnostic performance between CT and initial real-time RT-PCR for clinically suspected 2019 coronavirus disease (COVID-19) patients outside Wuhan, China,” Respiratory Medicinej,jul 2020 vol. 168, no. April, p. 105980, 2020, National library of medicine .,

14. Hsin-Jui Chen, Shanq-Jang Ruan, Sha-Wo Huang and Yan-Tsung Peng, “Lung X-ray Segmentation using Deep Convolutional Neural Networks on Contrast-Enhanced Binarized Images”, Multidisciplinary Digital Publishing Institute , Mathematics, April 2020.

15. Ranjani.R, DR.R.Priya, Efficient segmentation and classification of lung cancer Diagnosis Techniques Using Ct Images: A Review., Turkish Journal of Computer and Mathematics Education Vol.12 No.10 (2021), 2433-2440

16. Ranjani.R, Dr.R.Priya, FILTERING AND ENHANCEMENT OF LUNG COMPUTERIZED TOMOGRAPHY IMAGES FOR THE DIAGNOSIS OF COVID-19 AND LUNG CANCER, Design Engineering, Vol 2021: Issue 07, 13900-13918.

17. Ranjani.R, DR.R.Priya, **A** COMPARISON OF IMAGE RESTORATION TECHNIQUES OF LUNG CT COVID-19 IMAGES, International journal of Mechanical Engineering, Kalahari journals,Vol.6,3 Dec,2021,pg no, 15921593,issn:0974-5823.

18. . Jianning Chi, Yifei Zhang, Xiaosheng Yu, Ying Wang and Chengdong Wu,“Computed Tomography (CT) Image Quality Enhancement via a Uniform Framework Integrating Noise Estimation and Super-Resolution Networks”, Multidisciplinary Digital Publishing Institute , Sensors, July 2019.

19. Chamola, V.; Hassija, V.; Gupta, V.; Guizani, M. A Comprehensive Review of the COVID-19 Pandemic and the Role of IoT, Drones, AI, Blockchain, and 5G in Managing its Impact. *IEEE Access* **2020**, *8*, 90225–90265.

20. Bose, P.; Roy, S.; Ghosh, P. A Comparative NLP-Based Study on the Current Trends and Future Directions in COVID-19 Research. *IEEE Access* **2021**, *9*, 78341–78355. [

21. Mahase, E. Delta variant: What is Happening with Transmission, Hospital Admissions, and Restrictions? *BMJ* **2021**, *373*, n1513.